First H(z) Data Comparison of MAAT Structural-Selection Cosmology

Cosmic Chronometer Diagnostic and Chi-Square Scan

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Abstract

Paper 31 consolidated the MAAT dynamical-selection pipeline from structural defects to observable cosmological proxies. The present paper takes the next step and compares MAAT expansion histories directly with a compact Cosmic Chronometer $H(z)$ dataset.

Two diagnostics are performed. First, the representative MAAT v0.10 branch is compared to 31 chronometer measurements using the same $H_0 = 67.4$ normalisation as the reference ΛCDM curve. Second, a two-parameter scan over the kinetic scalar branch, $(\mu, \dot{\phi}_0)$, searches for stable MAAT trajectories with low chronometer chi-square.

The fixed v0.10 branch gives $\chi^2_{\mathrm{MAAT}} = 15.3661$ compared with $\chi^2_{\Lambda\mathrm{CDM}} = 14.8759$ for the reference model. The best stable scan point gives $\chi^2_{\mathrm{MAAT}} = 15.3674$, with $\mu = 15.8489$, $\dot{\phi}_0 = 2.7889$, $\max \Omega_{\mathrm{MAAT}} = 0.0267$, and maximum relative Hubble deviation 0.0235. For the two scanned MAAT parameters this corresponds to a diagnostic reduced chi-square $\chi^2_\nu \simeq 0.5299$ using $N - k = 29$ degrees of freedom. Thus MAAT is not favoured over ΛCDM in this first diagnostic comparison, but it remains dynamically stable, subdominant, and close to the reference chronometer fit.

The result should be read as a falsifiability bridge, not as evidence for a new cosmological model. It shows that the MAAT observable layer can be confronted with real expansion data using transparent and reproducible goodness-of-fit diagnostics.

1 Purpose and Status

The MAAT v0.10 observable layer produced expansion and growth proxies from a stable toy FLRW branch. Those observables were internally compared to a normalised ΛCDM reference, but no observational data entered the calculation.

The purpose of this paper is narrower and more empirical:

> *Can the MAAT expansion history be compared directly with observed Cosmic Chronometer $H(z)$ measurements without losing stability or becoming energetically dominant?*

This is not a precision cosmological analysis. No supernovae, BAO, CMB likelihoods, weak-lensing data, or redshift-space distortion likelihoods are included. The scan is a first diagnostic chronometer comparison and should be interpreted as a reproducibility test of the observable layer.

2 Chronometer Dataset and Reference Model

The benchmark uses a compact table of 31 Cosmic Chronometer measurements:

$$\{z_i, H_i, \sigma_i\}_{i=1}^{31}. \tag{1}$$

The reference model is a flat Planck-like ΛCDM curve with

$$H_0 = 67.4, \qquad \Omega_m = 0.315, \qquad \Omega_\Lambda = 0.685, \tag{2}$$

so that

$$H_{\Lambda\mathrm{CDM}}(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}. \tag{3}$$

The chi-square statistic is

$$\chi^2 = \sum_i \left(\frac{H_{\mathrm{model}}(z_i) - H_i}{\sigma_i}\right)^2. \tag{4}$$

When parameters are scanned, we also report the diagnostic reduced statistic

$$\chi^2_\nu = \frac{\chi^2}{N-k}, \tag{5}$$

where k is the number of scanned or fitted parameters. For the fixed-branch comparison no parameter is fit to the chronometer data ($k = 0$). For the two-parameter MAAT scan we use $k = 2$, corresponding to $(\mu, \dot{\phi}_0)$. This should not be confused with a full cosmological likelihood fit, since H_0, Ω_m, and the full covariance structure are held fixed.

In addition, the present diagnostic treats the chronometer errors as diagonal. Real Cosmic Chronometer analyses can involve stellar-population modelling systematics and correlated uncertainties. Those effects are not modelled here and are part of the next likelihood-level extension.

3 Fixed v0.10 Branch

The first test uses the representative v0.10 observable projection generated from the stable v0.9 branch. The dimensionless MAAT expansion history is converted to physical units by matching the same H_0 normalisation:

$$H_{\mathrm{MAAT}}(z) = H_0\frac{E_{\mathrm{MAAT}}(z)}{E_{\mathrm{MAAT}}(0)}. \tag{6}$$

Table 1: Fixed v0.10 branch compared with Cosmic Chronometer data.

Quantity	**Value**
Chronometer points	31
$\chi^2_{\Lambda\mathrm{CDM}}$	14.8759
χ^2_{MAAT}	15.3661
$\chi^2_{\Lambda\mathrm{CDM}}/N$	0.4799
χ^2_{MAAT}/N	0.4957
Fixed-branch $\chi^2_{\nu,\mathrm{MAAT}}$ $(k=0)$	0.4957
$\Delta\chi^2 = \chi^2_{\mathrm{MAAT}} - \chi^2_{\Lambda\mathrm{CDM}}$	0.4902
Maximum absolute pull, ΛCDM	1.5805
Maximum absolute pull, MAAT	1.6006

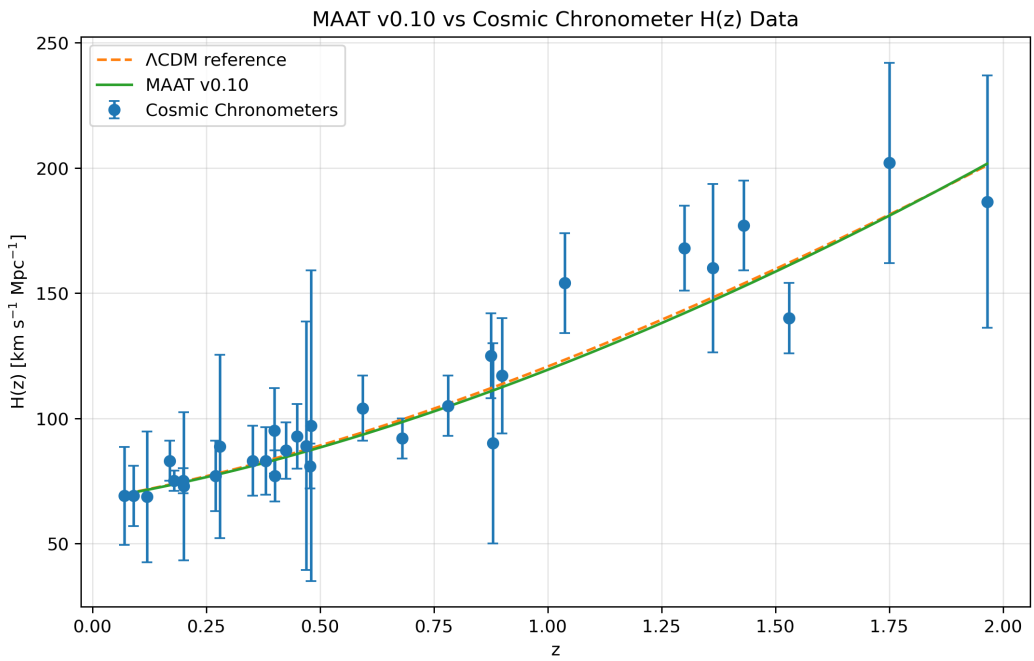

Figure 1: Fixed v0.10 MAAT branch compared with Cosmic Chronometer $H(z)$ data and the reference ΛCDM curve. The MAAT curve remains close to the reference but is not favoured over it by the simple chi-square statistic.

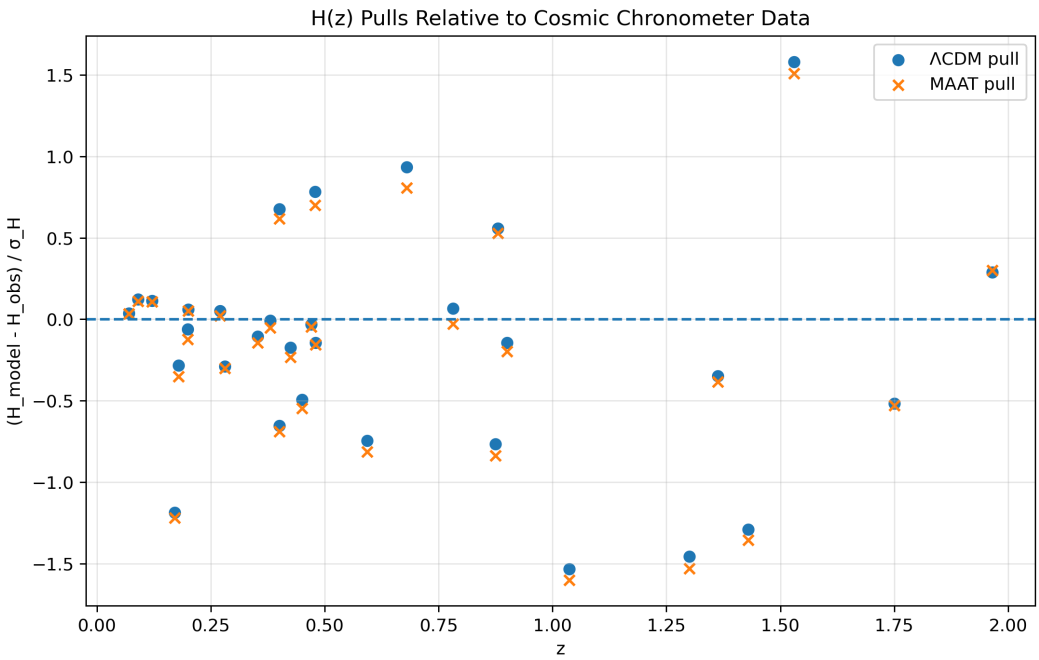

Figure 2: Chronometer pulls for the fixed v0.10 branch and the reference ΛCDM model. Both remain within comparable pull ranges in this diagnostic comparison.

4 Two-Parameter Chi-Square Scan

The second test scans the stable kinetic scalar branch over

$$\mu \in [10^{-2}, 10^{2.2}], \qquad \dot{\phi}_0 \in [0.15, 3.0], \tag{7}$$

using a 22×28 grid. Each trajectory is required to remain stable and to cover the full chronometer redshift range.

The stability cuts are

$$P_X > 0, \qquad c_s^2 > 0, \qquad \rho_{\mathrm{MAAT}} \geq 0. \tag{8}$$

Table 2: Two-parameter MAAT $H(z)$ chi-square scan.

Quantity	**Value**
Total scan points	616
Stable scan points	581
Degrees of freedom, $N - k$	29
Best μ	15.8489
Best $\dot{\phi}_0$	2.7889
Best χ^2_{MAAT}	15.3674
Best χ^2_{MAAT}/N	0.4957
Best reduced χ^2_ν $(k = 2)$	0.5299
$\Delta\chi^2$ relative to ΛCDM	0.4915
Best max Ω_{MAAT}	0.02668
Best max $\lvert\Delta H/H\rvert$	0.02347
Best min P_X	1.16209
Best min c_s^2	1.00026
Best max c_s^2	1.52630

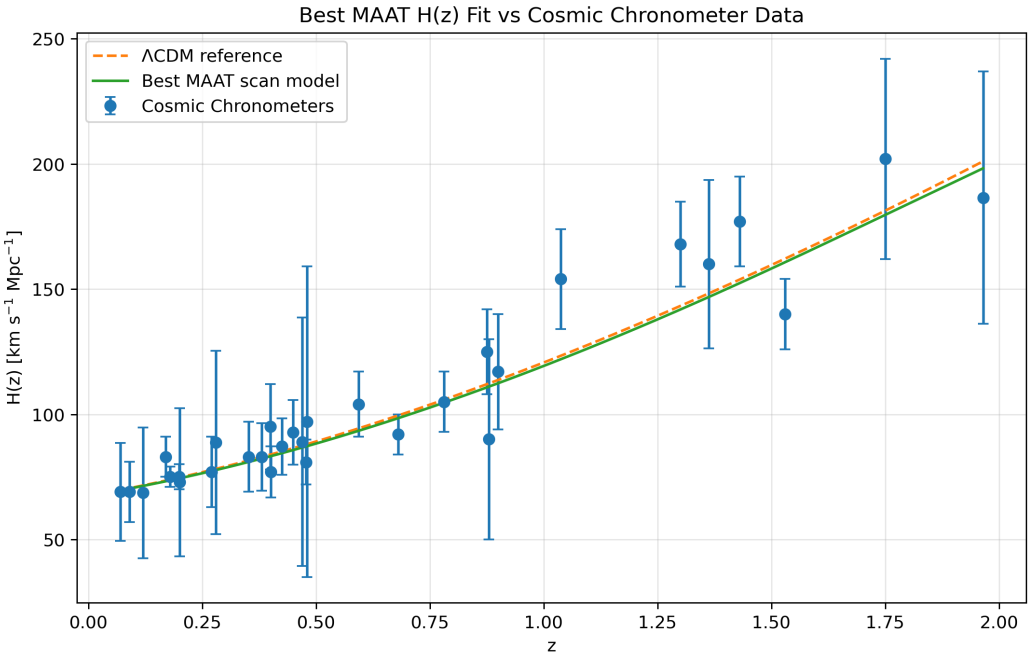

Figure 3: Best stable MAAT scan point compared with Cosmic Chronometer data. The best scan point remains close to ΛCDM but does not improve the chi-square.

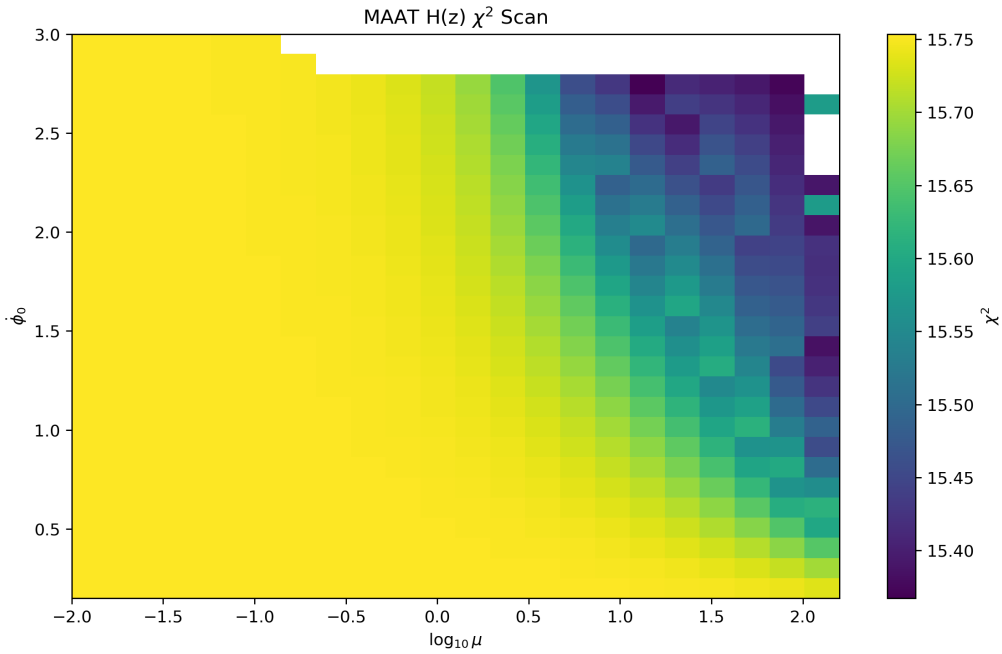

Figure 4: Chi-square landscape over the scanned $(\mu, \dot{\phi}_0)$ plane. Blank or high-cost regions correspond to unstable trajectories or poor chronometer agreement.

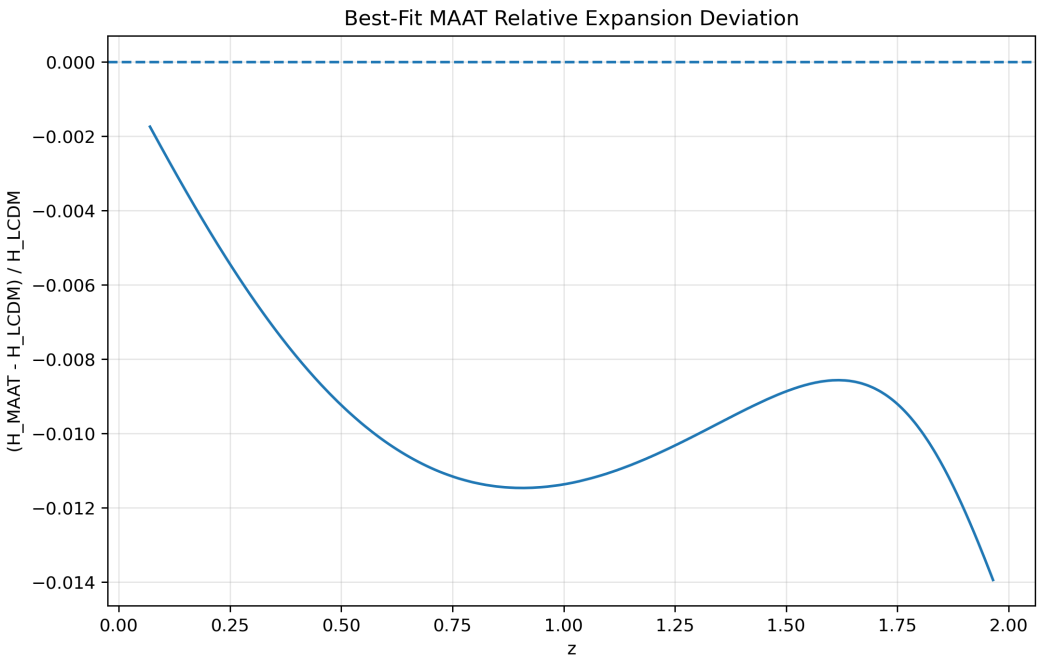

Figure 5: Relative expansion deviation of the best stable scan point. The best chronometer-compatible branch remains a small-deviation model.

5 Interpretation

The diagnostic result is deliberately modest:

MAAT is close to the chronometer reference fit, but not favoured over ΛCDM *in this first*

$H(z)$ comparison.

This is still useful. The branch is not excluded by this simple diagnostic, remains dynamically stable under the imposed scalar-sector cuts, and stays subdominant in the energy budget. Most importantly, the framework now admits a direct goodness-of-fit test against expansion data.

The physical reason for the close tracking is also instructive. Within the tested stable branch, the MAAT scalar sector remains energetically small, with $\max \Omega_{\mathrm{MAAT}} \simeq 0.027$, and therefore acts as a weak effective dark-energy perturbation rather than as a dominant new background component. The stability cuts select trajectories with positive kinetic response and positive sound-speed denominator, suppressing rapidly growing or strongly oscillatory deviations. In this limited sense the model behaves like a subdominant tracking sector: it can deform the expansion history at the few-percent level while remaining close to the ΛCDM background. This explains why the current branch is viable as a diagnostic test but does not yet produce a statistically preferred alternative.

The result also guards against overclaiming. If MAAT is to become a viable cosmological extension, it must eventually improve or explain data-level tensions without sacrificing stability. This paper does not show that. It only establishes the first transparent interface between the structural selection model and observational expansion data.

6 Limitations

- The chronometer table is compact and treated with diagonal errors only.
- H_0, Ω_m, and Ω_Λ are fixed rather than fit.
- No Bayesian evidence, covariance matrix, or full likelihood is computed.
- No supernova, BAO, CMB, weak-lensing, or redshift-space distortion data are included.
- The growth sector is not used in the chi-square scan.
- The MAAT model is still a toy scalar branch, not a complete cosmological model.

7 Reproducibility

All code and generated artifacts are stored in the Structural Selection repository:

https://github.com/Chris4081/structural-selection-principle

The relevant folder is:

`experiments/maat_hz_chi2_paper32/`

The main commands are:

```
python3 maat_hz_data_comparison_v01.py
python3 maat_hz_chi2_fit_v01.py
```

The scripts generate CSV/JSON files and figures in:

- `maat_hz_data_comparison_v01/`
- `maat_hz_chi2_fit_v01/`

Data and license note. The scripts embed a compact Cosmic Chronometer $H(z)$ table compiled from published literature values. No Hubble Space Telescope image/data products, DESI/BAO catalogues, CMB likelihoods, or weak-lensing datasets are bundled in this repository folder. The repository license applies only to the code, generated tables, and generated figures produced by the scripts. It does not relicense third-party observational measurements. The observational $H(z)$ values are treated as published scientific data and should be cited to the original Cosmic Chronometer publications and compilations when reused or discussed.

8 Conclusion

Paper 32 performs the first direct $H(z)$ data comparison of the MAAT structural-selection cosmology branch. The result is conservative but useful: MAAT does not outperform ΛCDM in this simple chronometer diagnostic, but it remains stable, subdominant, and close to the reference expansion fit.

The next step is not another qualitative extension. It is a full cosmological inference layer: parameter fitting, covariance-aware likelihoods, perturbation evolution, and joint comparison to chronometers, supernovae, BAO, CMB distance priors, redshift-space distortions, and weak-lensing data.

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